POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2018
Programme: BE | Full Marks: 100
Course: Chemistry | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Write the mechanism of basic buffer solution giving suitable example. 200ml of 0.1M acetic acid is mixed with 400ml of 0.2M sodium acetate solution. calculate the p^H of resulting mixture.(P^{Ka}=4.74) 7

   b) What do you mean by standard hydrogen electrode? How can we determine the standard electrode potential of zinc electrode by using SHE? From the given electrode potential values answer the following : 8

   i. Write the electrode reaction.

   ii. Write the cell reaction.

   iii. Calculate the e.m,f. of the cell at 27^0C when both the electrodes are coupled together .

   $E^0 Mg/Mg^{++} = +2.370$ V

   $E^0 Fe/Fe^{++} = +0.44$ V

   $R = 8.314$ J mol^{-1} K^{-1}

   F = 96500C

   $[Mg^{++}] = 0.1M$ $[Fe^{++}] = 0.01M$

2. a) Give reasons for the followings: 8

   i. $TiCl_3$ is coloured but $TiCl_4$ is colourless.

   ii. Transition elements show variable oxidation state.

   iii. Zn^{+2} salts are white.

   iv. Zn is not considered as transition element.

   b) Define ionization energy. Explain the factors affecting negativity. How electron affinity of chlorine is higher than Fluorine. 7

3. a) Define carbocations. How carbocations are stabilized? Write any two 7

methods of their formation.

b) What are elimination reactions? Write the mechanism and sterio chemistry of E_1 and E_2 reaction. 8

4. a) Write the mechanism of addition polymerization. 5

b) What is condensation polymerization? Write preparation, properties and uses of Nylon 6, 6. 5

c) Compare the properties of raw rubber and vulcanized rubber. 5

5. a) Write the principle and applications of mass spectroscopy. 8

b) What are lubricants? How are they classified? What are the applicants of lubrication? 7

6. a) What is water pollution? Discuss its causes, effects to human health and ways to control it. 8

b) What is hard water? How can it be measured in laboratory? 7

7. Write short notes on: (Any two) 2×5

a) Mechanism of rusting of iron

b) Nitration

c) Ozone layer depletion